

Internship Progress Report

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Internship Topic: Design and Development of Just-in-Time Immersive Video Coaching to Support CPAP Adherence

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1. Project Context

Design and implementation of a 5G Quality on Demand (QoD) enabled dynamic video generation pipeline. The project utilizes Edge Computing (MEC) via Raspberry Pi to detect real-time CPAP telemetry anomalies. Upon detection, it automatically orchestrates the generation and delivery of personalized, medically accurate coaching videos to patients under strict 5G/6G network delivery constraints (<60 seconds).

2. Summary of Achievements to Date

During this third month, the primary focus was on completing the end-to-end implementation of the edge-to-cloud pipeline for "Scenario 1" (existing asset retrieval), pre-generating the bilingual video library, and successfully executing the live demonstration in Berlin.

Asset Library Creation (Week 9 & 10)

- Manually generated a complete library of 15 high-quality coaching video assets using Google's generative video flow, alongside fully timestamped English and French subtitles (.en.vtt, .fr.vtt).



1_Mask_leak_adjust_straps.en.vtt	19/06/2026 15:31	VTT File	1 KB
1_Mask_leak_adjust_straps.fr.vtt	19/06/2026 15:31	VTT File	1 KB
2_Mask_leak_refit_while_lying_down.en.vtt	19/06/2026 15:31	VTT File	1 KB
2_Mask_leak_refit_while_lying_down.fr.vtt	19/06/2026 15:31	VTT File	1 KB
3_Cushion_cleaning.en.vtt	19/06/2026 15:31	VTT File	1 KB
3_Cushion_cleaning.fr.vtt	19/06/2026 15:31	VTT File	1 KB
4_Low_usage_ramp_mode.en.vtt	19/06/2026 15:31	VTT File	1 KB
4_Low_usage_ramp_mode.fr.vtt	19/06/2026 15:31	VTT File	1 KB
5_Mouth_breathing_use_chin_support.en...	19/06/2026 15:31	VTT File	1 KB
5_Mouth_breathing_use_chin_support.fr...	19/06/2026 15:31	VTT File	1 KB
6_Early_mask_removal.en.vtt	19/06/2026 15:31	VTT File	1 KB
6_Early_mask_removal.fr.vtt	19/06/2026 15:31	VTT File	1 KB
7_Nasal_congestion_relief.en.vtt	19/06/2026 15:31	VTT File	1 KB
7_Nasal_congestion_relief.fr.vtt	19/06/2026 15:31	VTT File	1 KB
8_Dry_mouth_use_humidifier.en.vtt	19/06/2026 15:31	VTT File	1 KB
8_Dry_mouth_use_humidifier.fr.vtt	19/06/2026 15:31	VTT File	1 KB
9_Daytime_desensitization.en.vtt	19/06/2026 15:31	VTT File	1 KB
9_Daytime_desensitization.fr.vtt	19/06/2026 15:31	VTT File	1 KB
10_Aerophagia_elevate_head.en.vtt	19/06/2026 15:31	VTT File	1 KB
10_Aerophagia_elevate_head.fr.vtt	19/06/2026 15:31	VTT File	1 KB
11_Positional_therapy_side_sleeping.en.vtt	19/06/2026 15:31	VTT File	1 KB
11_Positional_therapy_side_sleeping.fr.vtt	19/06/2026 15:31	VTT File	1 KB
12_High_breathing_events_contact_provi...	19/06/2026 15:31	VTT File	1 KB
12_High_breathing_events_contact_provi...	19/06/2026 15:31	VTT File	1 KB
13_Biomarker_changes_use_CPAP_and_c...	19/06/2026 15:31	VTT File	1 KB
13_Biomarker_changes_use_CPAP_and_c...	19/06/2026 15:31	VTT File	1 KB
14_Mask_style_change.en.vtt	19/06/2026 15:31	VTT File	1 KB
14_Mask_style_change.fr.vtt	19/06/2026 15:31	VTT File	1 KB
15_Contact_your_provider.en.vtt	19/06/2026 15:31	VTT File	1 KB
15_Contact_your_provider.fr.vtt	19/06/2026 15:31	VTT File	1 KB

- Deployed these assets to the Windows Virtual Machine server, successfully establishing the existing_assets repository required to validate Scenario 1 of the orchestration logic.

Edge Detection & Cloud Orchestration Deployment (Week 11)

- Developed and deployed the Python-based "watcher" script (cpap_edge_detector.py) on the Raspberry Pi edge node. The script successfully processes CPAP telemetry CSV drops in real-time, mapping clinical thresholds (e.g., Leaks ≥ 24 L/min, AHI ≥ 15) to the newly created 15-video matrix without manual intervention.
- Built and deployed the FastAPI orchestration server (main.py) on the Windows Virtual Machine. The API seamlessly receives edge triggers, fuzzy-matches the corresponding .mp4 and .vtt files, and executes a sub-second push to the central Web App Dashboard (159.84.143.151).

Berlin Demo Execution & Pipeline Validation (Week 12)

- Successfully demonstrated the fully integrated edge-to-cloud architecture (Scenario 1) at the Berlin Demo at the end of June 2026.





- Validated the dynamic catalogue population by writing a database migration script to safely inject generic maintenance videos (low relevance) into 20 test patient libraries, ensuring intrusive UI popups were preserved exclusively for acute, edge-detected clinical events.

3. Upcoming Schedule (July 2026)

This month, the focus will shift towards transitioning to "Scenario 2" (dynamic video generation), integrating the delayed machine learning components, and network simulation.

- **Week 13:** Integrate the delayed Keras anomaly-detection model. Convert it to TensorFlow Lite (TFLite) and deploy it on the Raspberry Pi, replacing the current Pandas-based threshold logic with the trained ML classifier.

- **Week 14:** Connect the pipeline to the active Google Vertex Veo API. This will enable Scenario 2, transitioning the system from retrieving pre-rendered assets to programmatically generating Image-to-Video animations in the cloud based on deterministic Python charts.
- **Week 15:** Implement a local mock CAMARA QoD API endpoint using Python to rigorously simulate and validate 5G/6G bandwidth reservation logic prior to physical network testing.
- **Week 16:** Conduct comprehensive latency benchmarking across the entire pipeline (from TFLite edge classification to Vertex API rendering to dashboard delivery).

4. Difficulties Encountered & Resolutions

Socket Binding Conflicts on Windows VM

- **Resolution:** During the deployment of the FastAPI orchestration server, an [Errno 10048] socket address conflict occurred, blocking port 8080. I resolved this by utilizing PowerShell network diagnostic tools (Get-NetTCPConnection) to isolate and terminate orphaned background Python processes, ensuring clean execution of the Uvicorn server.

Cross-Network Communication & Firewall Restrictions

- **Resolution:** The Raspberry Pi edge node initially failed to reach the Windows VM API (Connection timed out). I resolved this by explicitly configuring Windows Defender Firewall inbound TCP rules for the designated API ports and modifying the Uvicorn host binding (--host 0.0.0.0) to securely accept external cross-network POST requests.